

open circuit, thus triggering the alarm circuit connected to it.

Glass break sensors

Glass break detection can be accomplished in a variety of ways. The good old method is to bond narrow metal foil strips or threads to windows. These loops, when broken, open a circuit, which is detected by an alarm circuit. This mechanism is, however, very dif-



A glass break sensor

ficult to install and somewhat unsightly. A very popular means of glass break detection is to attach a small shock/impact sensor to the glass being protected. The tiny sensor has to be applied to the glass with a suction device or a special type of glue-tape. This device directly senses the vibration caused by the impact that causes the glass pane to break.

Recently, the technology has moved to detecting the acoustic sounds of breaking glass. Basically, it is an audio discriminator, triggered by the acoustic sound of breaking glass. This provides the advantage of one detector being able to cover a number of glass panes. The simpler designs of acoustic detectors sense the high-frequency tinkle of shattered glass.

Although glass break sensors that detect the sound of breaking glass are fairly reliable, some of these glass break detectors in a challenging environment may repeatedly trigger a false alarm. The detector sometimes mistakes high-pitched sounds with the sound of breaking glass (such as the sound of doorbell) and raises a false alert.

Today, glass break technology has been improved to eliminate false alarms from common occurrences by implementing 'dual-mode' detection tricks.

Dual-mode glass break detection

Glass break detectors that use dual-mode sensor technology (two sensors in one housing that operates by using two totally different technologies) were developed to eliminate false alarm problems in troubling environments. The dual-mode glass break detector works when both sensors are activated in a precise sequence, i.e., a compression wave is caused by the inward flex of a window just prior to breaking, followed by the acoustic sound of breaking glass. If no compression wave is detected, the sound of breaking glass alone will not trigger an alarm.

A dual-mode acoustic glass break detector unit listens to very low-frequency sound pressure wave—flex wave—created by the flexing of glass from a blow just before it breaks. Once a flex wave is detected, it enables a high-frequency detector to listen to the tinkle of the breaking glass. Since both technologies need to detect the glass breaking, false alarms are virtually eliminated.

This type of detector includes an acoustic transducer having a wide-band frequency response coupled to a dual-channel filter and signal processing circuit. A low-frequency channel detects an initial positive compression wave caused by the inward flex of the window and a high-frequency channel detects the acoustic spectrum, which is a characteristic of breaking glass. The two channels are combined in a logic circuit that is timed so that the low-frequency positive flex is detected initially with the high-frequency component following shortly thereafter. If both timing conditions are fulfilled an alarm is initiated. Additional circuitry is provided to inhibit the alarm if a negative compression wave is initially detected. The detection sequence is initiated by the loud sound of the breaking glass.

For fool-proof detection of the breaking of a window or glass pane, a suitable

acoustic transducer (such as a button microphone), a signal processing circuit responsive to the acoustic transducer for detecting a first low-frequency positive acoustic wave generated by an inward flex of the glass, and an alarm responsive to the signal processing circuit are necessary. Besides, the system must include a high-frequency bandpass filter for detecting high-frequency acoustic waves characteristic of breaking glass. Also, it must include a coincidence logic circuit that enables the alarm when the low-frequency acoustic wave is sensed during a predetermined time window that begins with a high-frequency event generated by the breaking glass. The alarm may then be triggered by sampling the high-frequency output of the transducer at a time after the initial time window.

The logic of this method takes advantage of the fact that the requisite high-frequency spectrum of acoustic waves will follow the initial positive low-frequency wave produced by the inward flex of the window or glass pane. An optional circuit may also be added to inhibit the alarm upon the detection of negative-going low frequency followed by high-frequency sounds. This inhibit feature significantly reduces the incidences of false alarms, specially those caused by a legitimate opening of the window.

Window alarm is an effective security measure that can deter most intruders. Today's advanced glass break detectors utilise highly evolved filtering algorithms to perform the complex task of determining the specific frequencies to analyse while ignoring similar sounds. Many of them have dual-mode glass break detector technology with excellent detection performance, employing microcontroller-based noise averaging technology. Noise averaging technique removes those unwanted background noises that might diminish the effectiveness of glass break detectors. ●

The author is an internationally certified freelance electronics circuit designer, technical writer, columnist, consultant, domain expert and trainer